ZECHARIAH DIEGELMAN

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Portfolio: zacdiegelman.github.io/Portfolio-Website/ | **LinkedIn:** linkedin.com/in/zechariahdiegelman

EDUCATION

University of North Carolina at Charlotte, Charlotte, NC — Dec 2027
Bachelor of Science in Mechanical Engineering | GPA: 3.20

Central Piedmont Community College, Charlotte, NC — Jul 2025
Associate in Engineering

TECHNICAL SKILLS

CAD / Design: SolidWorks, AutoCAD, nTop, Blender, parametric modeling, assemblies, engineering drawings, GD&T, dimensioning, BOM development, PrusaSlicer, Bambu Studio
Programming / Analysis: MATLAB, Mathcad, Python, C++, Arduino IDE, LabVIEW, Microsoft Excel, SolidWorks Simulation (FEA), nTop FEA
Electronics / Test: NI Multisim, National Instruments DAQ, strain gauges, Wheatstone bridges, sensors, instrumentation amplifiers
Fabrication / Tools: FDM 3D printing, prototype fabrication, mechanical assembly, automotive repair, woodworking machinery, hand and power tools, standard and metric measurement
Technical / Documentation: Git/GitHub, Microsoft Office, Google Workspace, technical report writing, engineering presentations

ENGINEERING PROJECTS

Multi-Axis Robotic Sorting Arm - PAP Robot, Junior Design - UNC Charlotte — Fall 2026 - Present

- Design and analyze the RA3 elbow subsystem for a four-axis robotic arm, including kinematics, free-body diagrams, actuator sizing, beam bending/deflection, load transfer, and factor-of-safety calculations.
- Develop SolidWorks components, assemblies, technical drawings, and BOMs while coordinating mechanical interfaces with the base, shoulder, wrist, and gripper subsystems.
- Selected by team members to lead CAD development and design iterations, coordinating subsystem assemblies and final CAD components across the team.

1995 Honda Accord Engine Swap & Mechanical Restoration, Personal Project — Feb 2026 - Present

- Independently completed an engine replacement and mechanical restoration involving drivetrain, clutch, braking, suspension, cooling, ignition, and supporting vehicle systems.
- Removed and installed the powertrain and serviced major components including the clutch, flywheel, timing system, water pump, axle, brakes, radiator, ignition components, suspension components, seals, hoses, and accessory drives.
- Diagnose mechanical issues, source replacement components, perform repairs and adjustments, and verify system operation through functional and road testing.
- Manage the project independently from vehicle acquisition through mechanical disassembly, parts sourcing, reassembly, troubleshooting, and road-ready completion.

Wi-Fi Controlled Rover, Solo Mechatronics II Project - UNC Charlotte — Summer 2026

- Independently designed and built a Wi-Fi-controlled rover using a microcontroller, DC motors, motor driver, battery system, and Arduino IDE.
- Implemented browser-based control and live camera streaming while integrating and debugging the mechanical, electrical, wireless, and software systems.

Drone Blade Guard (team project), Additive Manufacturing - UNC Charlotte — Spring 2026

- Modeled a lightweight racing-drone propeller guard in nTop with adjustable geometry, clearance, and manufacturing constraints; developed a reusable custom block for implicit-body conversion and material-saving geometry.

Additive Manufacturing Design & Benchmark Studies, 3D Printing Laboratory - UNC Charlotte — Fall 2025

- Created original SolidWorks designs, prepared prints with slicing software, evaluated print failures, revised geometry, and documented iterative design decisions and manufacturing lessons learned.
- Designed tolerance/fit artifacts and bridge/overhang benchmark tests to evaluate dimensional fit and unsupported-print limitations, then used printed outcomes to refine design-for-additive-manufacturing decisions.
- Created and iterated SolidWorks and nTop designs using tolerance and manufacturability evaluations and nTop FEA results to assess structural performance, print limitations, and geometry revisions.

PROFESSIONAL EXPERIENCE

Web Development & Business Operations Coordinator, J. Adam Builders - Charlotte, NC Jun 2026 - Sep 2026
- Developed and deployed the company's website and project portfolio while coordinating technical requirements, project content, deployment troubleshooting, and business needs directly with the owner.

Landscaper & Marketing Assistant, Barker Brothers LLC - Mint Hill, NC May 2023 - Jul 2024
- Constructed walkways, retaining walls, and landscape installations from field measurements using hand and power tools while adapting work to site conditions.
- Supported business operations through website development, photography, video production, and social media/digital marketing content.

Carpenter, Mint Hill Cabinetry - Monroe, NC Feb 2023 - May 2023
- Interpreted shop drawings, dimensions, and design requirements to fabricate and assemble high-end custom cabinetry using standard and metric measurements.
- Operated production equipment including table saws, dowel machines, edge banders, drills, and other shop tools; completed assembly, hardware installation, and fit-and-finish adjustments.

Lead Technician, Carolina's Grout Pros - Charlotte, NC Dec 2022 - Feb 2023
- Performed grout and tile cleaning/repair, caulking replacement, sealing, restoration, steam cleaning, and power-washing services while maintaining a dedicated work vehicle and equipment.

Crew Leader, LaborPanes - Charlotte, NC Aug 2021 - Jul 2022
- Supervised crew members on residential job sites, prepared job quotes, communicated with homeowners, and maintained vehicles and equipment while meeting quality and schedule expectations.

ADDITIONAL ENGINEERING & TECHNICAL PROJECTS

Mechanical Design & Analysis Project, Sophomore Design, UNC Charlotte Fall 2025
- Developed and evaluated a mechanical design using SolidWorks, GD&T/tolerancing, and FEA; produced engineering documentation and received Most Creative Bracket and Fan Favorite awards.

MATLAB Engineering Computation, Computational Methods - UNC Charlotte Summer 2026
- Developed scripts and functions to process engineering data, solve numerical problems, automate calculations, and verify numerical outputs with test cases.

DailyBread, and VerseKey iOS Apps, Independent Software/Product Project Summer-Fall 2026
- Independently designed, developed, tested, and published DailyBread to the Apple App Store while also developing VerseKey.
- Utilized React Native, Expo, and AI-assisted development tools to implement features, troubleshoot technical issues, and refine application functionality.
- Managed the software development lifecycle, including product design, testing, debugging, production builds, and App Store submission.

HONORS & AWARDS

Most Creative Bracket Award - MEGR 2151 Sophomore Design, UNC Charlotte Fall 2025
Fan Favorite Award - MEGR 2151 Sophomore Design, UNC Charlotte Fall 2025
Engineering Scholarship Recipient - Central Piedmont Community College Fall 2024 - Spring 2025
Dean's List - Central Piedmont Community College Spring 2025
Mu Alpha Theta National Mathematics Honor Society - Central Piedmont Community College

LEADERSHIP & COMMUNITY SERVICE

Mission & Community Service Volunteer, Costa Rica and Honduras 2010 - 2014
Participated in family mission and community-service work in Costa Rica (2010-2011) and with Forgotten Children Ministries in Honduras (2011-2014), supporting food distribution, residential construction, youth outreach, and assistance to underserved communities.
Developed conversational Spanish and cross-cultural communication skills through approximately four years of daily immersion while living in Central America.

Additional Community Service, Hickory Grove Baptist Church - Charlotte area, NC
Contributed to short-term local service projects including yard/property assistance and outreach at a community school/daycare.

LANGUAGES

Spanish - Conversational; developed through multi-year immersion in Costa Rica and Honduras.